HDOC Day-46

Aim: Write a C program to sort a variable-length integer array using Bubble Sort and pointers, then find the first and last occurrence of a target element.

C Program:

```
#include <stdio.h>

void Bubble_sort(int *a, int n) {
    int i, j, temp;
    for (i = 0; i < n - 1; i++) {
        for (j = 0; j < n - i - 1; j++) {
            if (*(a + j) > *(a + j + 1)) {
                temp = *(a + j);
                *(a + j) = *(a + j + 1);
                *(a + j + 1) = temp;
            }
        }
    }
}

void occurrence(int *a, int n) {
    int target, first = -1, last = -1, i;
    printf("Enter target: ");
    scanf("%d", &target);

    for (i = 0; i < n; i++) {
        if (*(a + i) == target) {
            if (first == -1)
                first = i;
            last = i;
        }
    }

    if (first == -1)
        printf("-1, -1\n");
    else {
        printf("First occurrence: %d at index %d\n",
               *(a + first), first);
        printf("Last occurrence: %d at index %d\n",
               *(a + last), last);
    }
}

int main() {
    int n, i;
    printf("Enter size of array: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Invalid size\n");
        return 0;
    }
    int nums[n];
    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &nums[i]);

    Bubble_sort(nums, n);
    printf("Sorted array: ");
```

```
    for (i = 0; i < n; i++)
        printf("%d ", nums[i]);
    printf("\n");
    occurrence(nums, n);
    return 0;
}
```

Sample Output 1:

```
Enter size of array: 6
Enter 6 elements: 5 7 7 8 8 10
Sorted array: 5 7 7 8 8 10
Enter target: 8
First occurrence: 8 at index 3
Last occurrence: 8 at index 4
```

Sample Output 2:

```
Enter size of array: 5
Enter 5 elements: 5 7 7 8 10
Sorted array: 5 7 7 8 10
Enter target: 8
First occurrence: 8 at index 3
Last occurrence: 8 at index 3
```

Sample Output 3:

```
Enter size of array: 6
Enter 6 elements: 5 7 7 8 8 10
Sorted array: 5 7 7 8 8 10
Enter target: 6
-1, -1
```